



*Internship at Tiramai.ai*

## **Bids Management**

*Submitted by:*

**Dhevesh Arun**

**PES1UG22CS182**

*Under the guidance of*

**Arka Basu**

Product Manager  
tiramai

**Internship duration: 22 weeks**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING  
FACULTY OF ENGINEERING  
PES UNIVERSITY**

(Established under Karnataka Act No. 16 of 2013)  
100ft Ring Road, Bengaluru – 560 085, Karnataka, India

## Joining Letter

**MARS**

driven to evolve

Dated: 01-Jan-2026

**Mr.Dhevesh Arun**

Email: Dhevesharun@gmail.com

Mobile: +91 8123323137

### Subject: Internship offer at Tiramai.ai

Dear Dhevesh,

We are pleased to offer you an **internship position at Tiramai.ai**, an AI-native organization under MARS Telecom Systems Pvt. Ltd. Your academic background and interest in AI engineering align well with the work we are doing to build next-generation, responsible AI platforms.

### Vision of the Organization

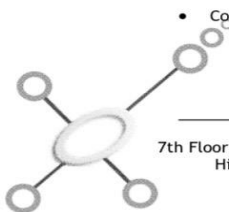
At Tiramai.ai, we envision a future where **responsible, human-centric AI** transforms how governments and mission-critical industries modernize their digital core. We are building a new breed of **AI-native platforms** — deeply verticalized, security-conscious, and purpose-built to solve the world's most complex challenges. From legacy modernization to intelligent verification systems, we deliver end-to-end GenAI solutions that empower enterprises and public systems to act faster, smarter, and with greater trust. Your internship will provide you with hands-on exposure to this mission.

### Why You Should Join Us

At Tiramai.ai, you will be part of a pioneering team redefining how AI integrates into enterprise ecosystems. You'll work alongside top AI architects and product leaders, developing solutions at the intersection of responsibility, scalability, and innovation.

We foster a culture that values:

- Purpose: Building AI that empowers, not replaces.
- Growth: Continuous learning and exposure to next-generation AI technologies.
- Integrity: Ethical AI, transparency, and accountability in every decision.
- Collaboration: Flat, innovation-driven teams that value your ideas and initiative.



### MARS Telecom Systems Private Limited

7th Floor, H-07, aVance Business Hub, M/s Phoenix Info City Limited, Survey No. 30, 35 & 38,  
HITECH City-2, Gachibowli, Hyderabad - 500 081, Ranga Reddy Dist., Telangana  
Tel: +91 40 6768 5555 / 5556 Fax: +91 40 6768 5594

## DECLARATION

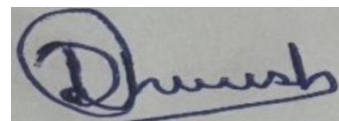
I hereby declare that the project entitled “**Bids Management**” has been carried out at *Tiramai.ai* by me under the guidance of **Arka Basu** and submitted in partial fulfillment of the credits for the degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester **22 week** duration. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

**PES1UG22CS1**

**82**

**Dhevesh**

**Arun**



## **ACKNOWLEDGEMENT**

**I would like to express my gratitude to my guide Arka Basu, Product Manager tiramai.ai for their continuous guidance, assistance and encouragement throughout the development of this project.**

**I am grateful to the internship coordinator Prof Mahitha G and Dr Bhargavi M, Dept. of Computer Science and Engineering , PES University for organizing, managing and helping out with the entire process.**

**I take this opportunity to thank Dr. Shylaja S S, Chairperson, Department of Computer Science and Engineering, PES University, for all the knowledge and support I have received from the department.**

**I am deeply grateful to Prof Jawahar Doreswamy, Chancellor, PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University for providing me various opportunities and enlightenment every step of the way.**

**Finally, this internship could not have been completed without the continual support and encouragement I have received from my parents and my friends.**

## **ABSTRACT**

I completed 16 weeks of my 22-week internship at Tiramai as an AI Engineering Intern, where I worked on the development of a Bid Management System for Mars under a partnership initiative. The primary objective of the project was to simplify and automate tender document analysis using AI and machine learning techniques. The system was designed to extract important information from tender documents, assist in bid evaluation, and predict the probability of winning bids, thereby reducing manual effort and improving efficiency. During the internship, I contributed to both frontend and backend development. I developed dashboards and user interfaces for tender management, analytics, reporting, audit, and settings modules using React-Vite, while also supporting the migration from Next.js to React-Vite for improved performance. On the backend, I worked with FastAPI, Milvus embeddings, RAG pipelines, Redis, WebSockets, Celery, and Docker. The internship enhanced my skills in AI integration, scalable application development, and full-stack engineering.

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## **1.Introduction**

The rapid growth of artificial intelligence and modern software technologies has transformed the way organizations manage large amounts of data and automate business processes. Many companies today are adopting AI-driven solutions to improve efficiency, reduce manual work, and support better decision-making. One such area is bid and tender management, where organizations must analyze large tender documents, manage bidding workflows, and make strategic decisions within limited timeframes.

This internship was carried out at Tiramai as an AI Engineering Intern, where work was done on the development of a Bid Management System for Mars as part of a partnership initiative. The main objective of the project was to simplify and automate the analysis of tender documents using artificial intelligence and machine learning techniques. The system was designed to extract important information from tender documents, assist in managing the bidding process, and predict the probability of winning bids. This helps organizations reduce the time and effort required for manual document analysis while improving efficiency and decision-making.

During the internship, contributions were made to both frontend and backend development of the application. The frontend was developed using React and Vite for building dashboards and user interfaces related to tender management, analytics, audit, reporting, and system settings. On the backend side, FastAPI was used for API development, while Milvus was used for vector embeddings and Retrieval-Augmented Generation (RAG) pipelines. Technologies such as Redis, Celery, WebSockets, and Docker were also used for real-time communication, asynchronous task processing, and deployment.

The internship provided practical exposure to enterprise-level software development, AI integration, scalable application design, and collaborative development workflows. It also helped in improving technical knowledge, problem-solving abilities, and teamwork skills in a professional working environment.

## **2. Company's Introduction**

Tiramai.ai is an AI-native organization operating under MARS Telecom Systems Pvt. Ltd., focused on developing responsible and human-centric artificial intelligence solutions for enterprise and mission-critical industries. The organization specializes in building AI-powered platforms that help modernize digital systems through automation, intelligent data processing, and scalable AI technologies. Tiramai.ai aims to deliver secure, reliable, and domain-specific AI solutions that support governments, enterprises, and large-scale organizations in improving operational efficiency and decision-making processes.

The company focuses on areas such as Generative AI, machine learning, intelligent verification systems, enterprise automation, and AI-driven digital transformation. Tiramai.ai emphasizes the development of scalable and ethical AI systems that integrate advanced technologies while maintaining transparency, accountability, and security standards.

The organization promotes a collaborative and innovation-driven work culture that encourages continuous learning, teamwork, and technical growth. During the internship, the company provided exposure to enterprise application development, AI engineering workflows, cloud-based tools, backend systems, and modern frontend technologies. Working at Tiramai.ai provided valuable industry experience in developing real-world AI-enabled applications and understanding professional software development practices.



### **3.Internship Project Details**

#### **3.1 Project Title**

Bids Management System

#### **3.2 Is it Part of Another Bigger Project?**

Yes, the project was part of a larger enterprise initiative focused on developing AI-powered solutions for intelligent document processing and workflow automation. The system was designed to support organizations in managing and analyzing tender and bid-related documents efficiently using modern AI technologies.

#### **3.3 Roles and Responsibilities**

During the internship, I worked as an AI Engineering Intern and contributed to both frontend and backend development activities of the application. My responsibilities included developing and enhancing user interfaces for modules such as tender management, analytics, reporting, audit tracking, and system settings using React-Vite. I also supported the migration from Next.js to React-Vite to improve application performance and maintainability.

On the backend side, I worked with FastAPI to develop REST APIs, create database tables, and manage application workflows. I contributed to AI-based functionalities by working with Retrieval-Augmented Generation (RAG) pipelines, embeddings, and vector databases using Milvus. Additionally, I worked with Redis, Celery, and WebSockets for caching, asynchronous task processing, and real-time communication features. Docker was used for containerization and deployment support. The internship also involved debugging, testing, collaborating with team members, attending review meetings, and maintaining proper technical documentation throughout the development process.

### **3.4 Project abstract and scope**

The project focused on the development of an AI-Based Bid Management System designed to simplify and automate the analysis of tender and bid-related documents. The primary objective of the system was to reduce the manual effort involved in reviewing large volumes of tender documents by using artificial intelligence and machine learning techniques for intelligent document processing and information extraction.

The application was designed to assist organizations in managing tender workflows, analyzing bid-related information, and predicting the probability of winning bids based on document analysis and contextual insights. The system included modules for tender management, reporting, audit tracking, analytics, and AI-driven document analysis. Advanced AI functionalities such as Retrieval-Augmented Generation (RAG) pipelines and vector embeddings were integrated to support contextual information retrieval and intelligent document understanding.

The scope of the project also included the development of scalable frontend and backend systems capable of handling enterprise-level workflows efficiently. The project aimed to improve operational efficiency, enhance document analysis accuracy, support better bid evaluation processes, and provide intelligent insights to assist organizations in making informed bidding decisions.

### **3.5 Technologies and Development Tools Used**

The project utilized various modern technologies and development tools for building a scalable AI-enabled enterprise application. React-Vite was used for frontend development to create responsive user interfaces and dashboards for modules such as tender management, analytics, reporting, and audit tracking. FastAPI was used on the backend for developing REST APIs, handling business logic, and managing communication between application services.

PostgreSQL was used as the primary relational database for storing and managing application data, while Milvus vector database was utilized for handling embeddings and similarity-based document retrieval. LangChain and Ollama were used for integrating Large Language Model (LLM) functionalities and implementing Retrieval-Augmented Generation (RAG) pipelines for intelligent document analysis and contextual information retrieval. RAG is an AI architecture that combines information retrieval with generative AI models to provide more accurate and context-aware responses.

Docker was used for containerization and deployment management, enabling consistent application environments across development and testing systems. Additional technologies such as Redis, Celery, and WebSockets were used to support caching, asynchronous task processing, and real-time communication features within the application.

### **3.6 Implementation Overview and Technical Contributions**

The implementation of the project involved the development and integration of multiple modules to support efficient bid and tender management workflows. On the frontend side, responsive dashboards and user interfaces were developed using React-Vite for modules such as tender management, analytics, reporting, audit tracking, and system settings. Efforts were also made to improve application performance and maintainability by supporting the migration from Next.js to React-Vite.

On the backend side, FastAPI was used for developing REST APIs, handling application workflows, and managing communication between frontend services and databases. PostgreSQL was used for relational data management, while Milvus vector database was integrated for embedding storage and similarity-based document retrieval functionalities.

AI-based capabilities were implemented using LangChain, Ollama, and Retrieval-Augmented Generation (RAG) pipelines to support intelligent document analysis and contextual information retrieval from tender documents. Redis, Celery, and WebSockets were used for caching, asynchronous task processing, and real-time communication features. Docker was utilized for containerization and deployment support to ensure consistent development and execution environments.

In addition to development activities, the internship involved debugging, testing, reviewing application workflows, participating in team discussions, and maintaining technical documentation throughout the project lifecycle.

### **3.7 Project results/Learning outcomes**

The internship provided valuable exposure to enterprise-level software development and AI-driven application development practices. Through the project, I gained practical experience in both frontend and backend development, including the design of responsive user interfaces, API development, database management, and scalable system integration.

I improved my understanding of modern web development technologies such as React-Vite and FastAPI, along with database systems including PostgreSQL and Milvus. The internship also helped me gain hands-on experience with AI technologies such as Large Language Models (LLMs), LangChain, Ollama, embeddings, and Retrieval-Augmented Generation (RAG) pipelines for intelligent document analysis and contextual information retrieval.

Additionally, I learned to work with tools and technologies such as Docker, Redis, Celery, and WebSockets for containerization, asynchronous task processing, caching, and real-time communication. The internship enhanced my problem-solving abilities, debugging skills, teamwork, communication, and understanding of professional software development workflows in an enterprise environment.

Overall, the internship significantly improved my technical knowledge, practical implementation skills, and confidence in developing scalable AI-enabled applications for real-world use cases.

#### **4. Conclusion**

The internship at Tiramai.ai as an AI Engineering Intern provided valuable industry exposure and practical experience in enterprise application development and artificial intelligence integration. Working on the AI-Based Bid Management System helped me understand real-world software development processes, scalable system design, and AI-driven document analysis techniques.

During the internship, I gained hands-on experience in frontend and backend development using technologies such as React-Vite, FastAPI, PostgreSQL, Milvus, LangChain, Ollama, Docker, Redis, and Celery. I also improved my understanding of Large Language Models (LLMs), embeddings, and Retrieval-Augmented Generation (RAG) pipelines for intelligent information retrieval and contextual analysis.

The internship enhanced my technical knowledge, problem-solving ability, teamwork, communication skills, and understanding of professional development workflows. Overall, the experience was highly beneficial in strengthening my skills in full-stack development, AI integration, and enterprise software engineering, while preparing me for future opportunities in the field of software and AI engineering.

## **5. References / Bibliography**

[1] Lewis, P., et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” Advances in Neural Information Processing Systems (NeurIPS), 2020. This paper introduces the concept of Retrieval-Augmented Generation (RAG), which combines parametric memory with external knowledge retrieval to improve performance in knowledge-intensive NLP tasks.

[2] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” Proceedings of NAACL-HLT, 2019. This work presents BERT, a transformer-based model that significantly improved natural language understanding through bidirectional training.

[3] Vaswani, A., et al., “Attention Is All You Need,” Advances in Neural Information Processing Systems (NeurIPS), 2017. This foundational paper introduced the Transformer architecture, which became the basis for modern large language models and attention-based systems.

[4] Reimers, N., & Gurevych, I., “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” Proceedings of EMNLP-IJCNLP, 2019. This paper presents Sentence-BERT, which enables efficient semantic similarity search using dense vector representations of sentences.

[5] OpenAI, “GPT-4 Technical Report,” OpenAI, 2023. This technical report describes the capabilities, architecture, and training methodology of GPT-4, a large-scale generative language model used for advanced natural language processing tasks.

[6] Milvus Team, “Milvus: A Cloud-Native Vector Database for Scalable Similarity Search,” 2023. This documentation explains Milvus, an open-source vector database designed for storing and searching high-dimensional embeddings efficiently in AI applications.

[7] Redis Labs, “Redis Documentation: In-Memory Data Structure Store,” 2024. This documentation explains Redis, an in-memory database used for caching, real-time data processing, and pub/sub messaging systems.

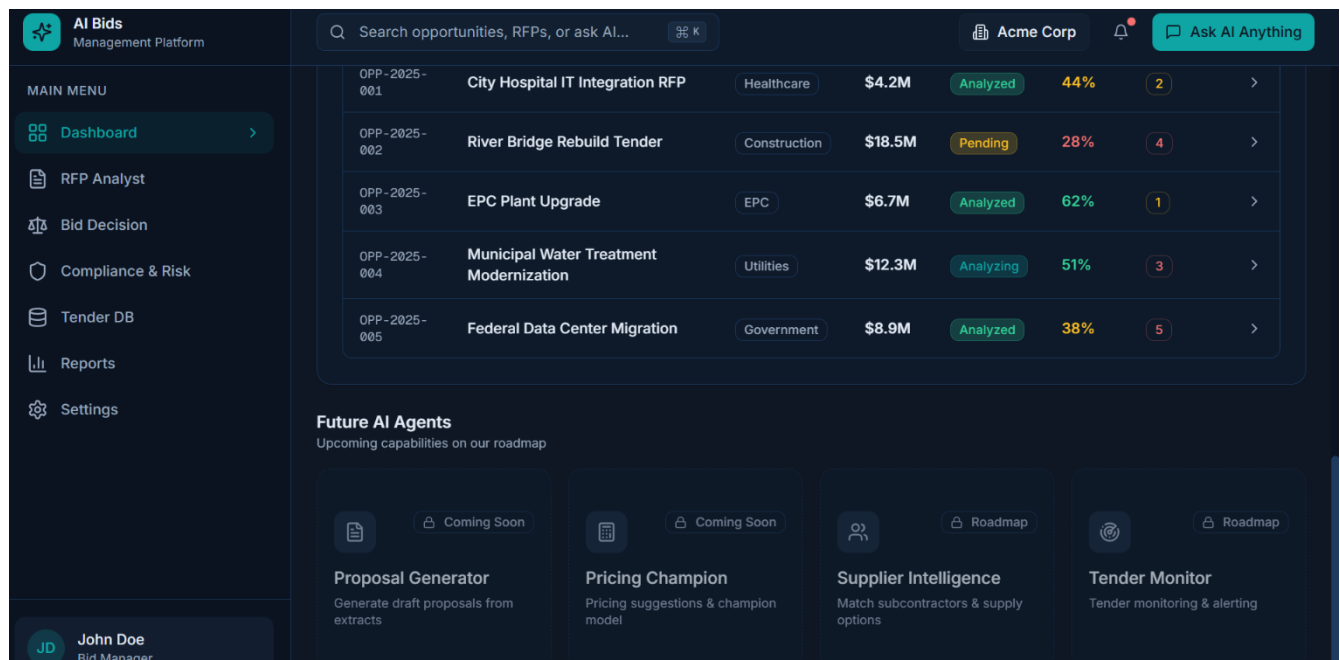
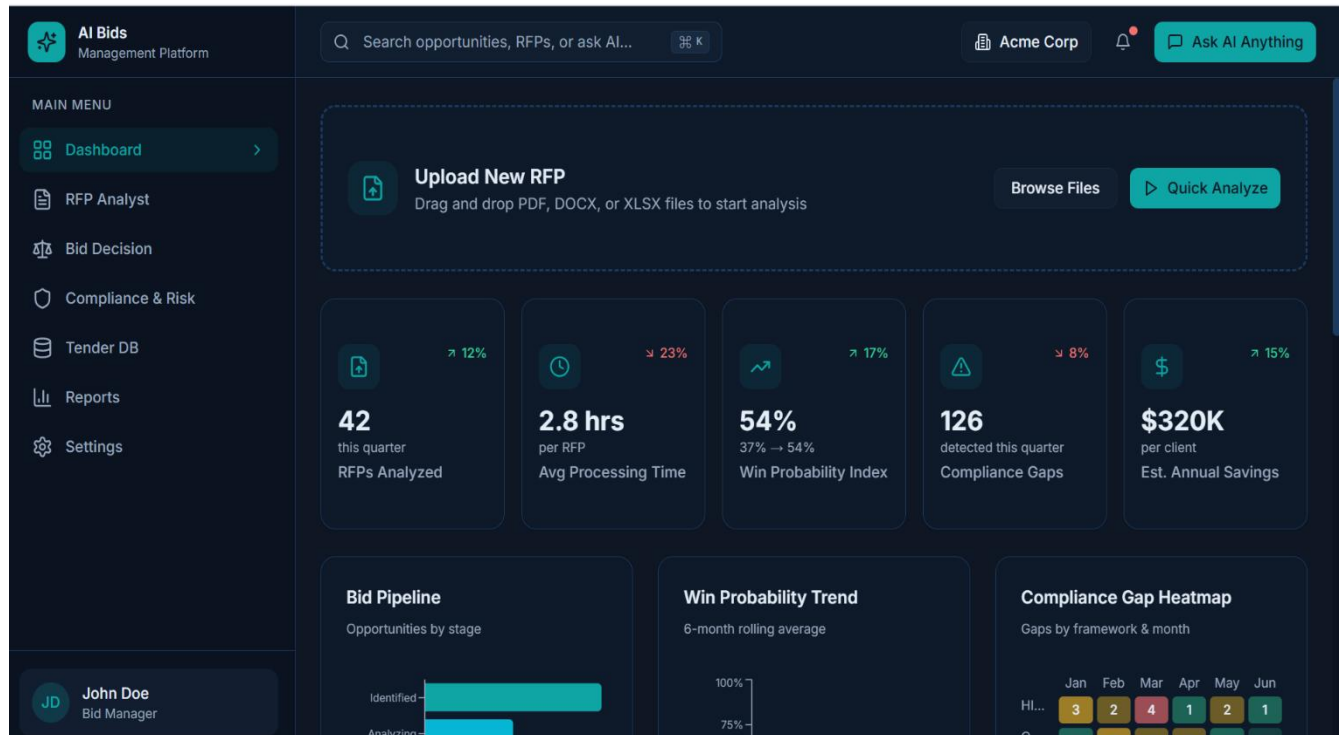
## **List of tables and figures**

### **List of tables**

- 1 Users
- 2 Documents
- 3 Tender\_Information
- 4 Analysis\_Results
- 5 AI/ML Models (Fitment Models)
- 6 Embedding Storage (Vector Database – Milvus)
- 7 Audit\_Logs
- 8 System\_Settings
- 9 Tender\_Categories
- 10 Notification\_Logs
- 11 Role\_Permissions
- 12 Activity\_Tracking
- 13 File Upload Metadata
- And more

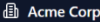


## List of figure




**AI Bids**  
Management Platform

K


3
Ask AI Anything

**MAIN MENU**

- Dashboard
- RFP Analyst
- Bid Decision**
- Compliance & Risk
- Tender DB
- Reports
- Settings


**John Doe**  
Bid Manager

← Back to Workspace

### Bid Decision Summary

City IT Modernization RFP

Ready for Submission

AI Confidence Score

**78%**

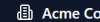
High confidence - Recommended to proceed

 Deal Overview

|                 |                   |
|-----------------|-------------------|
| Client          | Industry          |
| <b>MetroGov</b> | <b>Government</b> |
| Bid Value       | Deadline          |


**AI Bids**  
Management Platform

K


4
Ask AI Anything

**MAIN MENU**

- Dashboard
- RFP Analyst**
- Bid Decision
- Compliance & Risk
- Tender DB
- Reports
- Settings


**John Doe**  
Bid Manager

|                                     |                                                                                          |
|-------------------------------------|------------------------------------------------------------------------------------------|
| <b>Mobile Applications</b>          | iOS and Android native applications<br>Previous experience in similar mobile development |
| <b>Cloud Infrastructure</b>         | AWS/Azure cloud setup, configuration, and deployment                                     |
| <b>Security &amp; Compliance</b>    | SOC2, GDPR compliance implementation<br>May require external compliance audit partner    |
| <b>Testing &amp; QA</b>             | Comprehensive testing, automation, and quality assurance                                 |
| <b>Training &amp; Documentation</b> | User training, technical documentation, and knowledge transf                             |
| <b>Support &amp; Maintenance</b>    | 12 months of 24/7 support and maintenance                                                |
| <b>Project Management</b>           | Dedicated project management and coordination                                            |

Save Draft
Simulate Win Probability
Continue to Decision

**Timeline Constraints**  
Use agile methodology with bi-weekly sprints, dedicated resources

**Third-party Integration Dependencies**  
Early API documentation review, sandbox testing environment

**Resource Availability**  
Pre-allocated team confirmed, backup resources identified

**RATIONALE**

- Our platform development expertise aligns perfectly with 92% of the RFP requirements. We have 8+ similar projects delivered successfully.
- This contract presents an opportunity to establish presence in a growing market segment with potential for long-term partnership.
- SOC2 and GDPR compliance needs may require partnership with specialized compliance auditors, adding complexity and cost.

## RFP Analysis

RFP 2/10/2026

Win Probability  
**73%**



AI Confidence  
**85%**



Strategic Fit  
**95%**



Estimated Value  
**\$1.0M**



### Extracted Information

Client

**New Client**

Industry

**Healthcare**

Deadline

**Mar 3, 2026**

Contract Duration

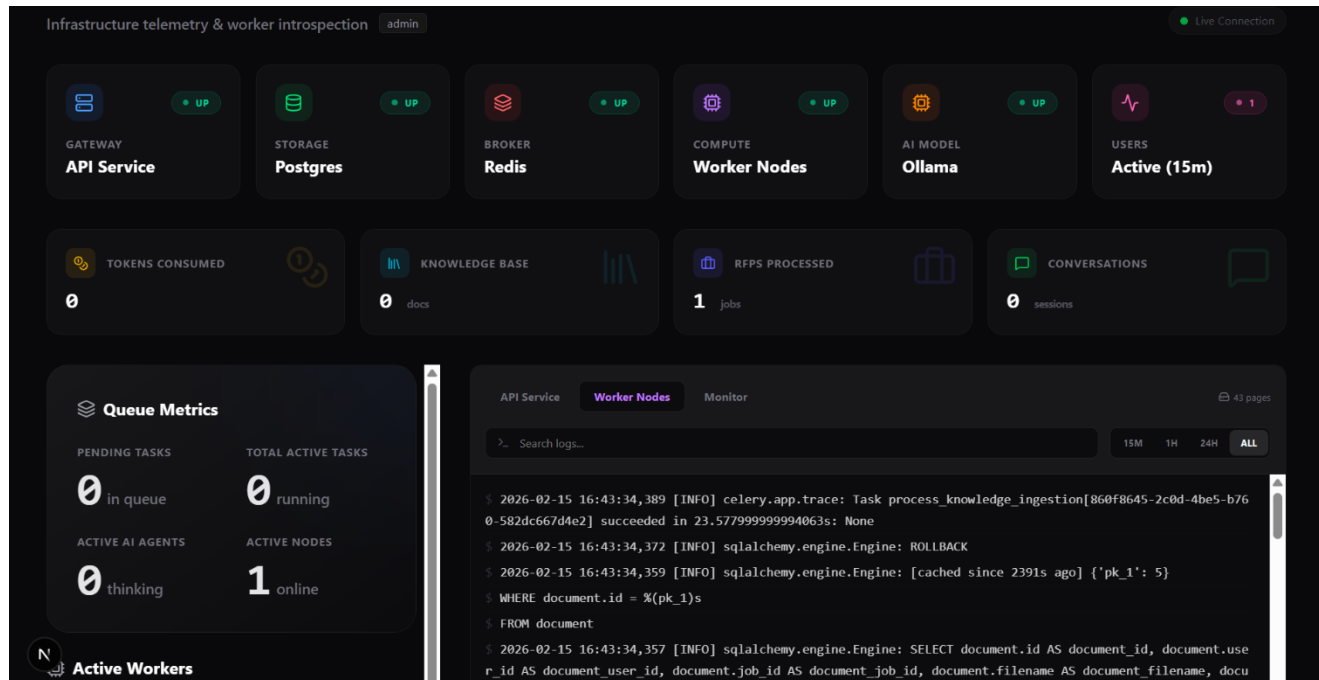
**12 months**

Description

AI-powered analysis of RFP 2/10/2026. This RFP requires

### Key Requirements

- ✓ Cloud infrastructure setup and migration
- ✓ RESTful API development and integration
- ✓ Native mobile applications (iOS/Android)
- ✓ Security compliance (SOC2, GDPR)
- ✓ Real-time analytics and reporting
- ✓ 24/7 technical support and maintenance
- ✓ Comprehensive documentation and training
- ✓ Agile development methodology



AI Bids

Management Platform

Back to Dashboard

Search opportunities, RFPs...

Acme Corp

Past Proposal

Upload

Dashboard

RFP Analyst

Knowledge Base

Compliance & Risk

Tender DB

Reports

Settings

admin

Bid Manager

Knowledge Base

AI-Powered

Add Knowledge Item

AI-powered searchable repository of organizational knowledge

q

×

All 17

Past Bids 4

Projects 4

Certifications 3

Policies 3

Case Studies 3

17 results for "c"

Clear search

Enterprise Cloud Infrastructure — Department of Defense

High Relevance

AI Relevance 95% Won

Past Bid

IT Infrastructure

Last updated: 2025-08-15

Cloud

DOD

Security

View Details

Use in Bid

View History

Cybersecurity Operations Center — DHS

High Relevance

AI Relevance 91% Won

Past Bid

Cybersecurity

Last updated: 2025-06-22

SOC

SIEM

FedRAMP

Compliance

View Details

Use in Bid

View History

AI Bids

Management Platform

Filters

Sync Source

Shortlisted Opportunities

Dashboard

RFP Analyst

Bid Decision

Knowledge Base

Compliance & Risk

Tender DB

Audit & Report

Settings

admin

Bid Manager

Tender Monitor

AI-POWERED

AI-powered tender discovery and opportunity screening

Total Tenders Pulled

554

+12% this month

New This Week

554

+5 from last week

Shortlisted

0

Active bids

High Value (>\$1M)

4

Total pipeline

Deadline < 7 days

134

Requires action

Search tenders by title, ID, or category...

All Sources

All Categories

All Values

| TITLE                                                               | PLATFORM | TENDER ID        | VALUE | DEADLINE   | CATEGORY      | ACTION      | VIEW   |
|---------------------------------------------------------------------|----------|------------------|-------|------------|---------------|-------------|--------|
| DOOR ASSEMBLY <span>NEW</span>                                      | SAM.gov  | 533a6412a8d64... | —     | 2026-03-16 | Manufacturing | ☆ Shortlist | 🔗 View |
| Y--SANTA ANA NWR Design-Build/CON Shade Str... <span>NEW</span>     | SAM.gov  | 5376a633a7c84... | —     | 2026-03-09 | Construction  | ☆ Shortlist | 🔗 View |
| TURBOCHARGER,EXHAUS <span>NEW</span>                                | SAM.gov  | 5419cad633364... | —     | 2026-05-01 | Manufacturing | ☆ Shortlist | 🔗 View |
| Construction Services of a Power Generation Mic... <span>NEW</span> | SAM.gov  | 54d74f8a9b7e4... | —     | 2026-03-11 | Construction  | ☆ Shortlist | 🔗 View |
| 6515--Hemodynamic Monitors/Platform <span>NEW</span>                | SAM.gov  | 55863716bc804... | —     | 2026-03-12 | Manufacturing | ☆ Shortlist | 🔗 View |

Showing 1-5 of 200 tenders

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3

4

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...

40

Next >

Auto-syncs every 14h

Dept. of CSE

January – April 2026

20